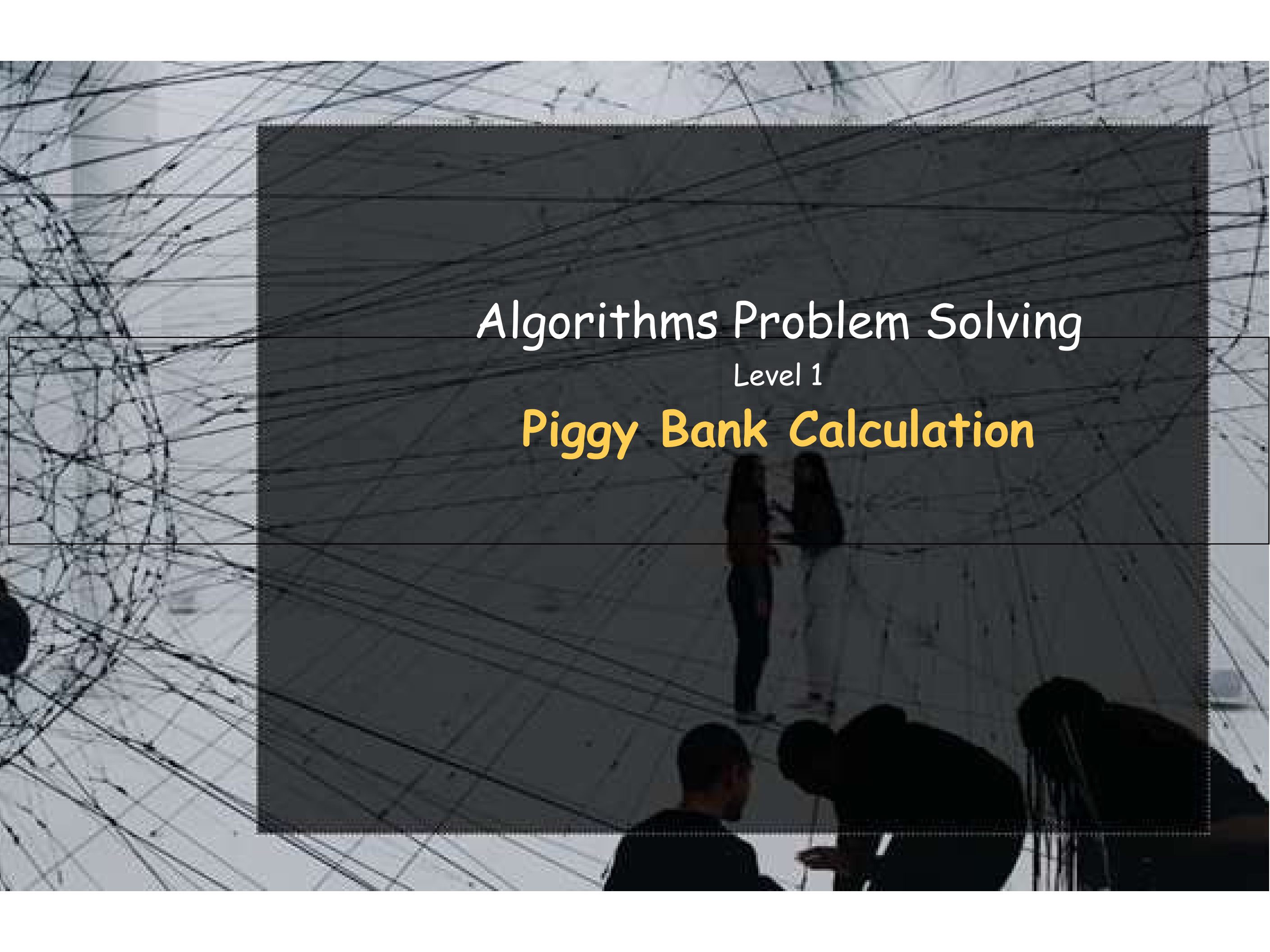


Algorithms Problem Solving

Level 1

Piggy Bank Calculation

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass ceiling that allows natural light to filter through, creating a pattern of light and shadow on the floor. In the foreground, the silhouettes of several people are visible, some standing and some sitting, looking up towards the ceiling. The overall atmosphere is one of architectural elegance and openness.

Algorithms Problem Solving

Level 1

Piggy Bank Calculation

Definition

- ◆ This program calculates the total value of coins and dollars stored in a piggy bank.
- ◆ It converts all amounts into pennies first, then converts the total into dollars.
- ◆ The program uses a `struct` to group the different types of money.

Explanation

- ◆ ``stPiggyBank`` is a structure that stores the number of Pennies, Nickels, Dimes, Quarters, and Dollars.
- ◆ ``readPiggyBank()`` creates a ``stPiggyBank`` object and reads the quantity of each type from the user.
- ◆ ``calculateTotalPennies()`` converts every value into pennies:
 - Pennies → 1 penny
 - Nickels → 5 pennies
 - Dimes → 10 pennies
 - Quarters → 25 pennies
 - Dollars → 100 pennies
- ◆ The converted values are added together to calculate ``TotalPennies``.
- ◆ ``printResult()`` converts pennies into dollars by dividing by ``100.0``.
- ◆ Using ``100.0`` instead of ``100`` makes the division produce a decimal result.
- ◆ ``printResult(calculateTotalPennies(readPiggyBank()))`` combines the functions:
- ◆ First, ``readPiggyBank()`` reads the data.
- ◆ Then, ``calculateTotalPennies()`` calculates the total.
- ◆ Finally, ``printResult()`` displays the result.

Steps in main()

- 1 ☐ Create the `stPiggyBank` structure.
- 2 ☐ Read the quantity of each coin and dollar.
- 3 ☐ Convert each value into pennies.
- 4 ☐ Add all penny values together.
- 5 ☐ Return the total number of pennies.
- 6 ☐ Divide the total pennies by `100.0` to get dollars.
- 7 ☐ Print both totals.

💡 Example

◆ If the user enters:

Pennies: 10

Nickels: 2

Dimes: 3

Quarters: 4

Dollars: 5

◆ The calculation is:

$$10 + (2 \times 5) + (3 \times 10) + (4 \times 25) + (5 \times 100)$$

◆ Total:

$$10 + 10 + 30 + 100 + 500 = 650 \text{ pennies}$$

➡ Result

The Total Pennies = 650

The Total Dollars = 6.5

Note

- ◆ All money is converted to the smallest unit, pennies, before calculating the total.
- ◆ ``100.0`` is used so that ``650 / 100.0`` produces ``6.5`` instead of integer division.

⊘ Warning

- ◆ The program does not validate whether the entered quantities are non-negative.
- ◆ It would technically accept negative quantities, even though they do not represent valid piggy bank contents.



Comparison

▨ Pennies-Based Calculation

- ✓ Converts everything to pennies first.
- ✓ Uses integers for the total value.
- ✓ Avoids losing fractional values during the main calculation.

▤ Direct Dollar Calculation

- ✓ Would require converting each coin directly to dollars.
- ✓ Would involve decimal values earlier.
- ✓ Can be less convenient for exact money calculations.

✦ Summary

- ✓ A `struct` groups all piggy bank contents together.
- ✓ Different coins are converted into pennies using their values.
- ✓ The total is calculated in pennies first.
- ✓ Dividing by `100.0` converts pennies into dollars with decimal precision.
- ✓ Functions separate input, calculation, and output.

Thank you 😊